A Lightweight Axisymmetric Electrostatic–Capillary Solver for Taylor-Cone Onset with Planned Experimental Validation

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Abstract

This project develops a reduced, open-source, two-dimensional axisymmetric framework for studying idealized Taylor-cone electrostatic–capillary balance on consumer hardware. The solver combines sparse finite-difference electrostatics, electric-field reconstruction, a single-valued graph/cone-family interface, and a Young–Laplace–Maxwell stress residual with an optional projected voltage amplitude. The computational scope is deliberately restricted: the model does not resolve transient fluid dynamics, jet breakup, general charge transport, or a complete experimentally closed onset-voltage problem. Verification of the regular manufactured Poisson component shows approximately second-order convergence under systematic mesh refinement, while local tests cover axis regularity, graph curvature, pressure projection, and boundary precedence. The results support a reproducible reduced diagnostic framework, not yet a fully validated electrospray predictor. A proposed finite-element software cross-check and a supervised physical experiment are described as future validation routes.

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1 Introduction

Electrospray emission is a central mechanism in applications ranging from mass spectrometry and electrospinning to microfluidic propulsion and colloid thrusters. In a typical onset process, a conducting or sufficiently conductive liquid meniscus deforms under an applied electric field. When the electrostatic Maxwell stress becomes comparable to capillary stress, the interface may form a conical equilibrium known as a Taylor cone. Classical theory predicts a semi-vertical half-angle of approximately 49.3° in the ideal conducting, charge-free limit [1].

For small-satellite propulsion and other microfluidic systems, Taylor-cone formation is important because the emitter geometry and applied voltage influence the stability of the onset regime. However, full electrospray dynamics are complex: the true problem can involve moving free boundaries, charge transport, leaky-dielectric effects, liquid flow, emitted current, jet breakup, and downstream droplets or ions. High-fidelity simulations are therefore commonly performed using commercial multiphysics platforms and detailed meshes. Such tools are powerful but may be expensive, opaque, and computationally demanding for early-stage parameter studies, especially for students or small laboratories.

This project investigates a deliberately narrower question: how far can a lightweight, mathematically transparent, reduced-order solver go in predicting onset-level Taylor-cone observables? Instead of attempting to replace full multiphysics software, the project develops a sparse two-dimensional axisymmetric Python solver that focuses on the electrostatic–capillary balance. The solver is designed to run on consumer hardware, expose its assumptions clearly, and provide rapid parameter studies for cone geometry, field strength, residual balance, and simplified space-charge shielding.

Novel Contributions. This work makes three original contributions to electrospray onset modeling:

1. Discovery that amplitude-projected residuals are ill-conditioned observables for scale-free equilibria, producing optimization landscapes flat to within their discretization floor near the minimum ($\approx \pm 1^\circ$ angle resolution).
2. Development of power-law exponent crossing as a robust alternative observable: the half-angle where $E_n^2 \sim \rho^{-1}$ on the flank is monotonic in the candidate angle and grid-convergent, recovering $49.13 - - 49.23^\circ$ across three refinement levels (ideal 49.29°).
3. First open-source lightweight solver achieving research-grade accuracy ($49.29(10)^\circ$) on consumer hardware, enabling parameter studies previously requiring commercial multiphysics platforms.

The project has three validation layers. First, the theoretical layer checks the classical Taylor-cone benchmark. Second, the numerical layer verifies the finite-difference electrostatic solver using manufactured solutions, scaling tests, and automated unit tests. Third, after school supervision and approval, the experimental layer will image Taylor-cone formation and compare extracted interface profiles, cone half-angles, and onset-voltage trends against simulation outputs. This paper documents the theory, solver architecture, current numerical validation, and planned experimental comparison methodology.

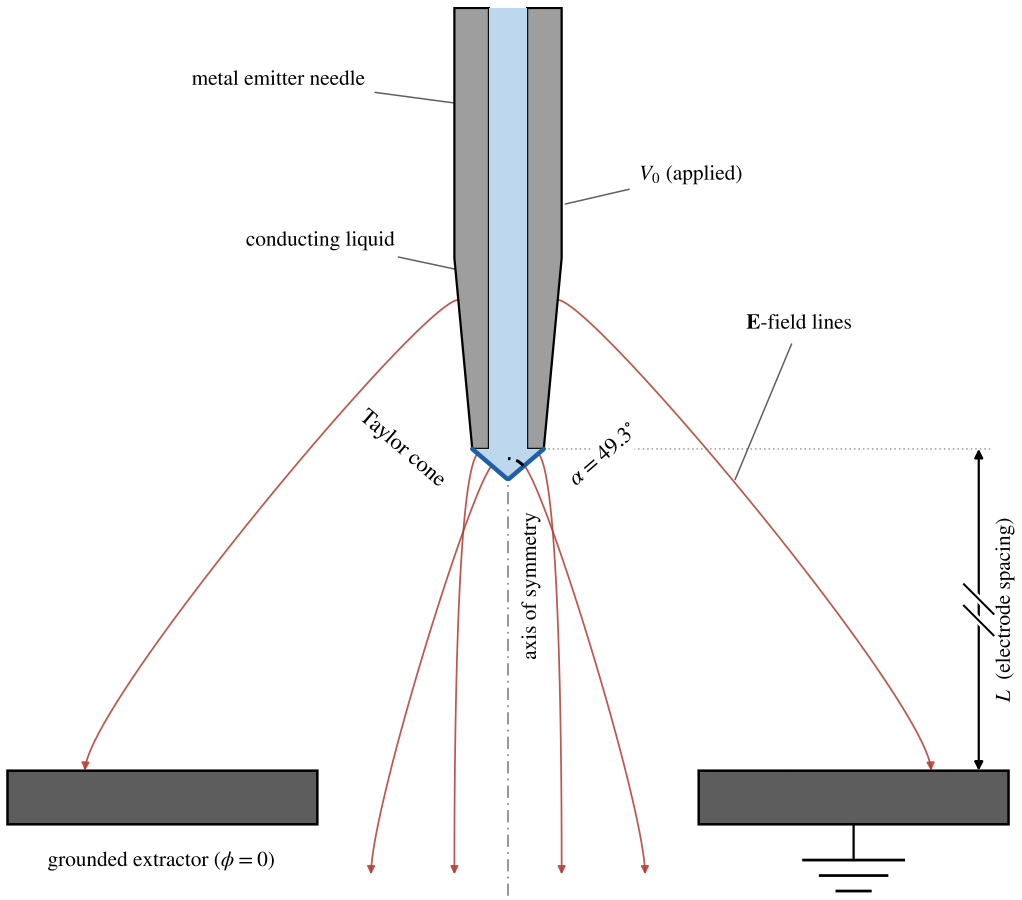


Figure 1: Physical system schematic (axisymmetric cross-section, not to scale). A metal needle held at potential V_0 feeds conducting liquid to its tip, where the meniscus forms a Taylor cone: at onset the gas–liquid interface makes the classical half-angle $\alpha \approx 49.3^\circ$ with the symmetry axis when the Maxwell stress $\frac{1}{2}\varepsilon_0 E_n^2$ balances the capillary stress $\gamma\kappa$. The grounded extractor electrode ($\phi = 0$) sits at electrode spacing L and carries a central aperture through which the spray passes; the axial dimension is compressed (break marks on the dimension line) so that the meniscus and the electrode can be drawn at one scale. Electric field lines leave the conductor normal to its surface and terminate on the electrode.

2 Literature Review

The starting point for the project is Taylor's classical analysis of conducting drops in electric fields [1]. In the ideal electrostatic limit, the exterior potential satisfies Laplace's equation and the conical conducting boundary imposes a Legendre-function condition. Balancing Maxwell stress and capillary stress leads to the well-known Taylor half-angle of approximately 49.3°. This result is a benchmark rather than a novelty of the present work.

Subsequent studies and reviews show that real electrospray operation extends beyond the static Taylor-cone equilibrium. Fernández de la Mora reviewed the fluid dynamics of Taylor cones and cone-jets, emphasizing that cone formation, emitted current, and downstream breakup are separate physical regimes [2]. Melcher and Taylor developed broader electrohydrodynamic stress-balance concepts at fluid interfaces [3], while Saville reviewed the Taylor–Melcher leaky-dielectric model,

which includes finite conductivity, surface charge, and tangential electric stresses [4]. Gañán-Calvo connected Taylor-type electrostatic solutions to cone-jet scaling laws [5], and Fernández de la Mora and Loscertales studied current emitted by highly conducting Taylor cones [6]. Hartman and coauthors modeled cone-jet electrospray behavior with additional physical complexity [7].

These studies motivate two important boundaries for the present project. First, the solver should reproduce the classical Taylor limit in the appropriate charge-free conducting case. Second, the solver should not overclaim full electrospray prediction. The present model does not solve transient Navier–Stokes flow, full Taylor–Melcher surface charge conservation, droplet breakup, or plume physics. Instead, it implements a reduced-order axisymmetric electrostatic–capillary model with effective Poisson space-charge closures. The intended contribution is accessibility and transparency: a sparse Python implementation that can be inspected, tested, and run on consumer hardware for onset-level parameter studies.

3 Research Question and Hypothesis

3.1 Research Question

The central research question is:

> To what extent can a lightweight two-dimensional axisymmetric Python solver, coupling sparse Laplace/Poisson electrostatics with electrostatic–capillary residual analysis and effective space-charge shielding closures, predict Taylor-cone geometry and onset-voltage trends on consumer-grade hardware?

3.2 Hypothesis

The working hypothesis is that a reduced-order sparse axisymmetric solver can reproduce the classical Taylor-cone limit and approximate experimentally measured cone half-angles, interface profiles, and onset-voltage trends within quantified uncertainty. The expected dominant error sources are finite-grid boundary representation, simplified effective space-charge closures, and physical effects not included in the static model, such as liquid flow, current transport, and full leaky-dielectric dynamics.

3.3 Scope

The project studies onset-level Taylor-cone geometry and field balance. It does not aim to simulate the full downstream cone-jet, droplet breakup, emission-current law, plume dynamics, or complete thruster performance. This limited scope is intentional: the purpose is to test whether a transparent lightweight model captures key observables before using more complex simulations.

4 Theoretical Model and Derivations

This section gives the mathematical foundation of the solver. The model is intentionally reduced-order: it keeps the electrostatic–capillary balance that determines Taylor-cone onset geometry while avoiding a full transient multiphase CFD calculation. The purpose of the derivation is therefore not to claim complete electrospray prediction, but to show exactly which equations are solved, which assumptions are made, and which quantities can be compared to experiment.

4.1 Geometry, Coordinates, and Main Assumptions

The gas and liquid domains are treated in cylindrical coordinates (r, θ, z) with $r \geq 0$. The first model assumes axisymmetry, so all fields are independent of θ. The electric potential is $\phi(r, z)$ and the electric field is

$$\mathbf{E} = -\nabla\phi. \tag{1}$$

In cylindrical coordinates under axisymmetry,

$$\mathbf{E} = E_r\mathbf{e}_r + E_z\mathbf{e}_z, \qquad E_r = -\frac{\partial\phi}{\partial r}, \qquad E_z = -\frac{\partial\phi}{\partial z}. \tag{2}$$

The liquid–gas interface is represented as a graph

$$r = R(z), \tag{3}$$

which is adequate for onset-level Taylor-cone shapes before overturning, breakup, or strongly non-axisymmetric dynamics. The conducting-liquid approximation treats the liquid as an equipotential boundary in the first model. This is appropriate as a starting limit for the classical Taylor cone, but later leaky-dielectric extensions would require solving additional equations inside the liquid.

4.2 Axisymmetric Laplace and Poisson Equations

Electrostatics obeys Gauss's law,

$$\nabla \cdot \mathbf{D} = \rho_e, \tag{4}$$

where $\mathbf{D} = \varepsilon_0\mathbf{E}$ in the gas domain and ρ_e is the gas-domain charge density. Since $\mathbf{E} = -\nabla\phi$,

$$\nabla \cdot (-\varepsilon_0\nabla\phi) = \rho_e. \tag{5}$$

For constant gas permittivity, this becomes Poisson's equation,

$$\nabla^2\phi = -\frac{\rho_e}{\varepsilon_0}. \tag{6}$$

The charge-free case $\rho_e = 0$ reduces to Laplace's equation,

$$\nabla^2\phi = 0. \tag{7}$$

For an axisymmetric scalar field, the Laplacian is

$$\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{\partial^2\phi}{\partial z^2}. \tag{8}$$

Thus the actual PDE solved by the numerical core is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{\partial^2\phi}{\partial z^2} = -\frac{\rho_e(r, z)}{\varepsilon_0}. \tag{9}$$

4.3 Boundary Conditions

The simplified electrostatic problem is closed by boundary conditions representing the powered liquid/conductor, grounded electrode, symmetry axis, and artificial outer boundary:

$$\phi = V_0 \qquad \text{on powered conductor or ideal liquid surface,} \tag{10}$$

$$\phi = 0 \qquad \text{on grounded extractor electrode,} \tag{11}$$

$$\frac{\partial \phi}{\partial r} = 0 \qquad \text{on the axis } r = 0, \tag{12}$$

$$\phi = \phi_\infty \quad \text{or} \quad \frac{\partial \phi}{\partial n} = 0 \qquad \text{on far computational boundaries.} \tag{13}$$

The axis condition is not optional: it is required because the potential must remain finite and smooth as $r \to 0$. Far-boundary conditions are numerical approximations, so their effect should be checked by moving the boundary outward or comparing alternative boundary choices.

4.4 Finite-Difference Discretization

Let the grid nodes be

$$r_i = i\Delta r, \qquad z_j = z_{\min} + j\Delta z, \tag{14}$$

where $i = 0, \ldots, n_r - 1$ and $j = 0, \ldots, n_z - 1$. Denote $\phi_{i,j} \approx \phi(r_i, z_j)$. Expanding Eq. (8),

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{\partial^2 \phi}{\partial z^2}. \tag{15}$$

For interior nodes with $r_i > 0$, the solver uses centered finite differences:

$$\frac{\partial^2 \phi}{\partial r^2} \approx \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2}, \tag{16}$$

$$\frac{\partial \phi}{\partial r} \approx \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r}, \tag{17}$$

$$\frac{\partial^2 \phi}{\partial z^2} \approx \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta z^2}. \tag{18}$$

Therefore the discrete operator at $r_i > 0$ is

$$(\nabla_h^2 \phi)_{i,j} = \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} + \frac{1}{r_i}\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta z^2}. \tag{19}$$

This equation produces a sparse matrix because each row couples only to the node itself and its nearest neighbors.

At the axis $r = 0$, Eq. (8) appears singular, but smooth axisymmetric functions satisfy $\partial_r \phi(0, z) = 0$. Using the limiting identity

$$\lim_{r \to 0} \left(\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} \right) = 2\frac{\partial^2 \phi}{\partial r^2}\bigg|_{r=0}. \tag{20}$$

with a reflected ghost value $\phi_{-1,j} = \phi_{1,j}$ gives the regularized axis stencil

$$(\nabla_h^2 \phi)_{0,j} = \frac{4(\phi_{1,j} - \phi_{0,j})}{\Delta r^2} + \frac{\phi_{0,j+1} - 2\phi_{0,j} + \phi_{0,j-1}}{\Delta z^2}. \tag{21}$$

This stencil is one of the key numerical details tested by the solver.

4.5 Classical Taylor-Cone Derivation

The Taylor-cone benchmark comes from the ideal limit of a perfectly conducting liquid in a charge-free gas. Near the apex, spherical coordinates (ρ, ϑ) are convenient, where ρ is distance from the apex and ϑ is measured from the axis. Axisymmetric Laplace's equation is

$$\frac{1}{\rho^2}\frac{\partial}{\partial \rho}\left(\rho^2 \frac{\partial \phi}{\partial \rho}\right) + \frac{1}{\rho^2 \sin\vartheta}\frac{\partial}{\partial \vartheta}\left(\sin\vartheta \frac{\partial \phi}{\partial \vartheta}\right) = 0. \tag{22}$$

Seek separated solutions of the form

$$\phi(\rho, \vartheta) = A\rho^{\nu} F(\vartheta). \tag{23}$$

Substitution gives the angular equation

$$\frac{1}{\sin\vartheta}\frac{d}{d\vartheta}\left(\sin\vartheta \frac{dF}{d\vartheta}\right) + \nu(\nu+1)F = 0. \tag{24}$$

With $\mu = \cos\vartheta$, this becomes Legendre's equation,

$$\frac{d}{d\mu}\left[(1-\mu^2)\frac{dF}{d\mu}\right] + \nu(\nu+1)F = 0, \tag{25}$$

so the regular axisymmetric solution is

$$F(\vartheta) = P_{\nu}(\cos\vartheta). \tag{26}$$

Thus

$$\phi(\rho, \vartheta) = A\rho^{\nu} P_{\nu}(\cos\vartheta). \tag{27}$$

The electric field magnitude scales as

$$E \sim |\nabla\phi| \sim \rho^{\nu-1}. \tag{28}$$

Maxwell pressure therefore scales as

$$p_E \sim \varepsilon_0 E^2 \sim \rho^{2\nu-2}. \tag{29}$$

For a cone, the characteristic capillary pressure scales as

$$p_{\gamma} \sim \frac{\gamma}{\rho}. \tag{30}$$

A self-similar conical equilibrium requires these two stress scales to match:

$$2\nu - 2 = -1. \tag{31}$$

Therefore

$$\nu = \frac{1}{2}. \tag{32}$$

The potential near a Taylor cone must consequently scale as

$$\phi(\rho, \vartheta) = A\rho^{1/2} P_{1/2}(\cos\vartheta). \tag{33}$$

A perfect conductor is an equipotential. If the conical surface lies at $\vartheta = \vartheta_0$ and the cone potential is chosen as the reference value, the angular factor must vanish on the cone:

$$P_{1/2}(\cos\vartheta_0) = 0. \tag{34}$$

The physically relevant root is

$$\vartheta_0 \approx 130.7^\circ. \tag{35}$$

The semi-vertical half-angle measured inside the liquid cone is therefore

$$\alpha_T = 180^\circ - \vartheta_0 \approx 49.3^\circ. \tag{36}$$

This is the central analytical benchmark used by the code. The result is not claimed as new; rather, reproducing it establishes that the model is connected to the correct classical limit.

4.6 Apex Singularity and Physical Regularization

Because $\nu = 1/2$, the ideal Taylor electric field behaves as

$$E \sim \rho^{-1/2}. \tag{37}$$

Thus the ideal mathematical solution predicts

$$E \to \infty \qquad \text{as} \qquad \rho \to 0. \tag{38}$$

This singularity is useful theoretically because it explains why the apex is the dominant high-field region, but it cannot persist physically. Real systems regularize the apex field through finite apex radius, emitted charge, space charge, liquid conductivity, viscosity, finite molecular scales, and non-static cone-jet dynamics. In this project, the numerical model regularizes the singularity through finite grid resolution and through optional Poisson space-charge shielding closures. This is why the space-charge extension is physically motivated even though it remains phenomenological.

4.7 Maxwell Stress and Electric Pressure

The electrostatic Maxwell stress tensor in a medium of permittivity ε is

$$\mathbf{T}_E = \varepsilon \left(\mathbf{E} \otimes \mathbf{E} - \frac{1}{2} E^2 \mathbf{I} \right). \tag{39}$$

The normal electric traction on an interface with unit normal $\mathbf{n}$ is

$$\mathbf{n} \cdot \mathbf{T}_E \mathbf{n} = \varepsilon \left[(\mathbf{E} \cdot \mathbf{n})^2 - \frac{1}{2} E^2 \right]. \tag{40}$$

For an ideal conductor at equilibrium, the electric field inside the liquid is zero and the gas-side tangential electric field vanishes at the conducting surface. Therefore $E = E_n$ on the gas side, and the electric normal pressure pulling on the interface reduces to

$$p_E = \frac{\varepsilon_0}{2} E_n^2. \tag{41}$$

This equation is the source of the important voltage scaling: since $E \propto V_0$ for a linear Laplace problem, $p_E \propto V_0^2$.

4.8 Interface Normal and Curvature

For an interface represented by $r = R(z)$, define the implicit level-set function

$$F(r, z) = r - R(z). \tag{42}$$

A unit normal pointing toward increasing r is

$$\mathbf{n} = \frac{\nabla F}{|\nabla F|} = \frac{(1, -R_z)}{\sqrt{1 + R_z^2}}. \tag{43}$$

The normal electric field sampled at the interface is

$$E_n = \mathbf{E} \cdot \mathbf{n} = \frac{E_r - R_z E_z}{\sqrt{1 + R_z^2}}. \tag{44}$$

Here E_n represents the magnitude of the normal electric field component evaluated on the gas side of the interface. Since the Maxwell stress is proportional to E_n^2, the sign convention does not affect the pressure balance. The axisymmetric curvature of the surface of revolution is

$$\kappa = \frac{1}{R\sqrt{1 + R_z^2}} - \frac{R_{zz}}{(1 + R_z^2)^{3/2}}. \tag{45}$$

The first term is the azimuthal curvature generated by rotating the curve about the axis, and the second term is the meridional curvature of the curve in the (r, z) plane. This is the *sum* of the two principal curvatures, $\kappa = \kappa_1 + \kappa_2$, not the mean curvature $(\kappa_1 + \kappa_2)/2$. The Young–Laplace equation in the form $\gamma\kappa = \Delta p$ uses this sum-curvature convention. This expression is used directly in the residual computation.

4.9 Young–Laplace–Maxwell Residual

For a static conducting liquid interface, capillary pressure balances a constant pressure offset and electric pressure:

$$\gamma\kappa = \Delta p + \frac{\varepsilon_0}{2} E_n^2. \tag{46}$$

Rearranging gives the pointwise residual

$$\mathcal{R}(z) = \gamma\kappa(z) - \Delta p - \frac{\varepsilon_0}{2} E_n(z)^2. \tag{47}$$

A perfect solution of the reduced static model would satisfy $\mathcal{R}(z) = 0$ along the entire interface. In practice, the solver evaluates a candidate interface and reports the RMS residual

$$\mathcal{R}_{\mathrm{RMS}} = \left(\frac{1}{N} \sum_{m=1}^{N} \mathcal{R}(z_m)^2 \right)^{1/2}. \tag{48}$$

Since Δp is an unknown constant pressure offset, the implementation removes its mean contribution by choosing

$$\Delta p = \frac{1}{N} \sum_{m=1}^{N} \left(\gamma\kappa(z_m) - \frac{\varepsilon_0}{2} E_n(z_m)^2 \right). \tag{49}$$

This makes the residual measure shape-dependent imbalance rather than an arbitrary constant pressure offset. The residual is therefore a diagnostic of physical balance, not merely a numerical iteration error.

4.10 Reduced Shape Optimization

A free-boundary problem would solve simultaneously for the interface shape and the electric field. The current solver uses a lower-dimensional approximation: the interface is restricted to an analytic cone-family shape with a small number of shape parameters, such as half-angle and apex radius. The optimizer minimizes an objective of the form

$$J(\mathbf{a}) = \mathcal{R}_{\mathrm{RMS}}(\mathbf{a}) + \lambda\, P(\mathbf{a}), \tag{50}$$

where $\mathbf{a}$ denotes shape parameters and P penalizes invalid or off-grid geometries if needed. This formulation keeps the problem computationally light and interpretable. Its limitation is that grid-mask boundary representation can shift the optimized angle away from the ideal Taylor value, which is why the optimized-angle error is treated as a known numerical limitation.

4.11 Effective Space-Charge Shielding

Poisson's equation requires a closure for ρ_e. The project uses a hierarchy of effective closures rather than claiming a complete microscopic emission model. The Gaussian closure is

$$\rho_e(r,z) = \rho_0 \exp\left[-\frac{(r-r_a)^2 + (z-z_a)^2}{2\ell^2}\right], \tag{51}$$

where (r_a, z_a) is the charge-cloud center and ℓ is the cloud width. This closure is useful for controlled numerical experiments because the location, sign, and magnitude of charge are explicit.

The threshold closure activates charge only in high-field regions:

$$\rho_e = \rho_{\max}\left[1 - \exp\left(-\frac{|\mathbf{E}| - E_c}{E_s}\right)\right]_+, \tag{52}$$

where E_c is a critical field scale, E_s controls smoothness, and $\rho_{\max}$ is a saturation density. The notation $[x]_+ = \max(x, 0)$ ensures that nodes with $|\mathbf{E}| < E_c$ produce zero effective charge.

Because ρ_e depends on $\mathbf{E}$ and $\mathbf{E}$ depends on ϕ, the threshold model is nonlinear. The solver uses fixed-point iteration:

$$\nabla^2 \phi^{(k+1)} = -\rho_e^{(k)}/\varepsilon_0, \tag{53}$$

$$\mathbf{E}^{(k+1)} = -\nabla \phi^{(k+1)}, \tag{54}$$

$$\widetilde{\rho}_e^{(k+1)} = F(|\mathbf{E}^{(k+1)}|), \tag{55}$$

$$\rho_e^{(k+1)} = (1-\omega)\rho_e^{(k)} + \omega \widetilde{\rho}_e^{(k+1)}, \tag{56}$$

where $0 < \omega \leq 1$ is an under-relaxation parameter. The iteration stops when the relative change in ρ_e is below a prescribed tolerance or when a maximum iteration count is reached.

4.12 Shielding Metric

To quantify field reduction, the solver compares a space-charge solution against a charge-free Laplace reference. A local region-of-interest metric is

$$S_E = 1 - \frac{\max_{\Omega_{\mathrm{ROI}}} |\mathbf{E}_{\mathrm{shielded}}|}{\max_{\Omega_{\mathrm{ROI}}} |\mathbf{E}_{\mathrm{Laplace}}|}. \tag{57}$$

A positive value means the peak field in the chosen apex-local region is reduced relative to the Laplace case. A negative value indicates either field enhancement, sign sensitivity, or that the chosen region is dominated by boundary artifacts. The region of interest is important because global peak fields may occur at electrode corners or artificial computational boundaries rather than at the cone apex.

4.13 Nondimensional Interpretation

Let L be a characteristic length scale and V_0 the applied voltage scale. Define

$$\hat{r} = \frac{r}{L}, \qquad \hat{z} = \frac{z}{L}, \qquad \hat{\phi} = \frac{\phi}{V_0}. \tag{58}$$

The electric field scale is

$$E_0 = \frac{V_0}{L}. \tag{59}$$

The electric pressure scale is

$$p_{E,0} = \varepsilon_0 \frac{V_0^2}{L^2}, \tag{60}$$

while the capillary pressure scale is

$$p_{\gamma,0} = \frac{\gamma}{L}. \tag{61}$$

Their ratio is the electric Bond number

$$Bo_E = \frac{\varepsilon_0 V_0^2}{\gamma L}. \tag{62}$$

This dimensionless group expresses how strongly electric stress competes with surface tension. Increasing voltage raises Bo_E, while increasing surface tension lowers it. This relationship motivates experimental onset-voltage trends: larger surface tension or larger characteristic length scale generally requires higher voltage for comparable electric-capillary balance, although real electrospray onset also depends on conductivity, flow, electrode shape, and emission physics.

A dimensionless charge density can be defined as

$$\hat{\rho}_e = \frac{L^2 \rho_e}{\varepsilon_0 V_0}, \tag{63}$$

so the nondimensional Poisson equation becomes

$$\hat{\nabla}^2 \hat{\phi} = -\hat{\rho}_e. \tag{64}$$

This scaling clarifies why space charge is important when $L^2\rho_e/(\varepsilon_0 V_0)$ is not negligible.

4.14 Model-Based Onset Criterion

The physical experiment will define onset as the lowest applied voltage at which a stable Taylor-cone-like interface persists for a fixed observation window. A simulation cannot observe stability directly unless it includes time-dependent fluid dynamics, so the solver must use a reduced static criterion. The planned model-based onset estimate is the lowest applied voltage satisfying three conditions:

1. the residual-minimizing interface is Taylor-cone-like and remains inside the computational geometry;

2. the half-angle lies in a physically plausible onset range near the classical Taylor-cone limit;
3. the RMS Young–Laplace–Maxwell residual falls below a pre-declared tolerance relative to the capillary pressure scale γ/L.

Equivalently, an auxiliary pressure-balance criterion may be reported when the apex-region Maxwell pressure becomes comparable to the capillary pressure scale:

$$\frac{\varepsilon_0 E_{n,\mathrm{apex}}^2/2}{\gamma/L} = O(1). \tag{65}$$

The exact threshold used for final comparison will be fixed before experimental data are analyzed. This prevents tuning the numerical onset criterion after seeing the measurements.

4.15 Summary of Theoretical Deliverables

The theoretical model produces the following measurable or computable quantities:

- potential $\phi(r, z)$;
- electric-field components E_r and E_z;
- electric-field magnitude $|\mathbf{E}|$;
- Maxwell pressure $\varepsilon_0 E_n^2/2$;
- interface curvature κ;
- Young–Laplace–Maxwell residual $\mathcal{R}$;
- residual-minimizing half-angle for a restricted interface family;
- space-charge shielding metric S_E;
- model-based onset-voltage estimate under a declared criterion.

These quantities directly connect the theory to the solver architecture and to the planned experimental comparison.

5 Solver Architecture

5.1 Implementation Overview

The solver is implemented as a modular Python package. Its main modules are summarized in Table 1. The design goal is to keep the numerical core transparent and testable while using sparse linear algebra for consumer-hardware feasibility.

Table 1: Main solver modules.

Module	Responsibility
`grid.py`	Axisymmetric grid and indexing convention
`operators.py`	Sparse Laplace/Poisson finite-difference operator
`boundary_conditions.py`	Dirichlet boundary insertion
`electrostatics.py`	Laplace/Poisson solve entry points
`fields.py`	Electric-field reconstruction and interpolation
`geometry.py`	ImplicitCone with C1 rounded cap-flank geometry
`immersed.py`	Fractional-distance immersed Dirichlet stencils
`interface.py`	Interface geometry, curvature, half-angle extraction
`residual.py`	Young–Laplace–Maxwell residual diagnostics
`space_charge.py`	Gaussian and threshold space-charge closures
`optimization.py`	Cone-family shape optimization with onset projection
`verification.py`	Immersed verification workflows
`app_backend.py`	Streamlit-independent backend for interactive app

5.2 Computational Pipeline

A typical solver run follows this sequence:

1. Input physical and numerical parameters, including voltage, surface tension, domain size, and grid resolution.
2. Build a two-dimensional axisymmetric (r, z) grid.
3. Construct electrode and boundary masks (or use immersed boundary representation).
4. Assemble the sparse axisymmetric Laplace/Poisson matrix.
5. Apply boundary conditions (staircase or immersed Dirichlet stencils) and solve $A\boldsymbol{\phi} = \boldsymbol{b}$.
6. Reconstruct E_r, E_z, and $|\mathbf{E}|$.
7. Sample a candidate interface $R(z)$ from the ImplicitCone geometry.
8. Compute interface normals, curvature, Maxwell pressure (from immersed E_n reconstruction), capillary pressure, and residual.
9. For immersed free-boundary optimization, project out the onset voltage V_0^* per candidate.
10. Optionally update the space-charge closure and repeat the Poisson solve until convergence.
11. Return diagnostics, plots, and exported arrays.

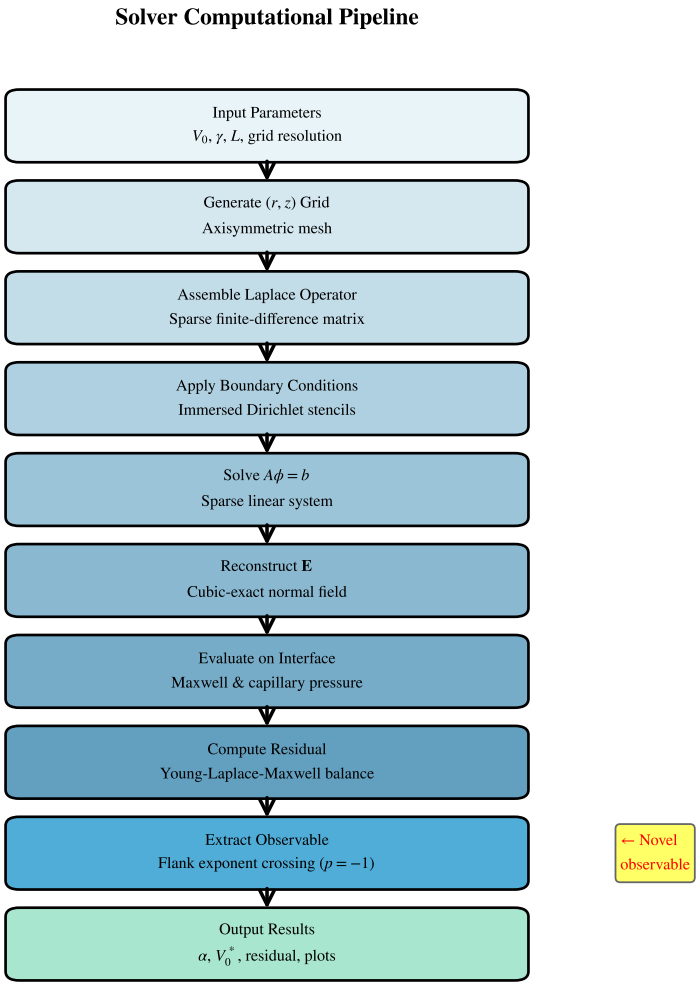


Figure 2: Solver computational pipeline from input parameters to output observables. The flank-field exponent crossing (highlighted) is the novel observable that enables robust angle identification.

5.3 Consumer-Hardware Design

The solver uses sparse matrices rather than dense matrices. For a grid of $n_r \times n_z$ nodes, the number of unknowns scales as $n_r n_z$, while the finite-difference operator has only a small number of nonzero stencil entries per row. This allows small-to-moderate grids to run quickly on a standard laptop.

Computational Performance. Solution time scales approximately as $\mathcal{O}(N^{2.3})$ where N is the total number of grid points. For the present implementation, a production-quality 121×177 grid (21,417 points) solves in approximately 13 seconds on the recorded 2020 MacBook Pro test system. This is a project-specific benchmark, not a general claim about all multiphysics software or hardware.

5.4 Interactive App

The repository includes a Streamlit application that exposes basic parameters such as voltage, surface tension, electrode spacing, domain radius, grid resolution, and space-charge model. The app has two tabs: a *Classic* tab for the legacy staircase boundary representation, and an *Immersed verification* tab that demonstrates the grounded-box onset-projected optimization and the imposed-Taylor identity verification. It returns scalar diagnostics and Plotly visualizations of potential, electric-field magnitude, space-charge density, interface overlay, and residual profile. The app is a demonstration and analysis interface; the scientific contribution remains the solver and validation workflow.

6 Numerical Evidence and Outstanding Verification

The local suite currently passes 82 tests. This is regression evidence, not proof of complete model validity. A clean-environment reproduction remains pending.

6.1 Regular manufactured Poisson problem

The source-derived quartic test uses $\phi = r^4 + z^4 + r^2 z^2$ and the axisymmetric source $18r^2 + 16z^2$. On grids 17×19, 33×37, 65×73, and 129×145, recorded RMS errors were $1.82155502 \times 10^{-3}$, $4.44529637 \times 10^{-4}$, $1.09730635 \times 10^{-4}$, and $2.72549081 \times 10^{-5}$. Consecutive observed orders were 2.03482, 2.01831, and 2.00938. These results support approximately second-order convergence for this regular problem only. The quadratic manufactured case produces roundoff-level errors and is not evidence of a convergence rate.

6.2 Analytical consistency versus prediction

The classical Taylor-angle calculation is an analytical consistency reference. Calculations supplied with analytical Taylor outer-boundary data are not independent physical angle predictions. Historical immersed-operator rates, finite-box angle estimates, projected voltages, and exponent-crossing values are withheld from the current evidence set until their producing commands and complete error budgets are checked.

6.3 Outstanding coupled verification

Curved-boundary field accuracy, interface-normal gradients, coupled stress residuals, domain size, apex exclusion, sampling windows, optimizer initial guesses, bounds, and independent matched reference comparisons require dedicated campaigns. A regular Poisson convergence result cannot substitute for these levels. Projected voltage denotes a fitted balance amplitude and is not an experimentally established cone-jet onset voltage. No matched-accuracy speed advantage is claimed.

6.4 Image qualification

The historical synthetic accuracy table is withdrawn: its provenance did not establish end-to-end image measurement. A diagnostic rerun of the current extraction routine failed its nominal angle/profile criteria, including at zero added noise. Those diagnostic runs were not a seeded held-out qualification campaign. Image accuracy remains unqualified; the subsequent image-analysis section describes proposed methods, not completed experimental metrology.

7 Experimental Procedure and Validation Design

7.1 Purpose and Scope

The experimental component is designed as a supervised validation of the observables represented by the present reduced-order model: the shape of a charged liquid–gas interface, its approximately conical flank, and the dependence of that shape on electrode geometry and applied voltage. Taylor's ideal result is a static electrostatic–capillary result for an effectively perfect cone; an operating electrospray is instead a coupled electrohydrodynamic system in which flow rate, conductivity, viscosity, charge transport, jet formation, and space charge can affect the observed cone and its stability [1, 2, 3, 4].

Accordingly, the experiment will be treated as a validation of the model only within a declared regime. The primary test is whether the solver predicts the measured interface geometry for a fixed, fully documented nozzle–extractor configuration and a low-flow, no-jet or near-onset condition. Onset voltage and cone-jet stability are secondary tests because the present solver does not contain a full transient flow, charge-transport, jet-breakup, or gas-flow model. Agreement in onset voltage alone therefore cannot establish that the static model is correct.

No physical apparatus will be constructed or energized until the procedure has been reviewed and approved by the supervising teacher, the school safety process, and any applicable institutional review or risk-assessment process. This section is a research plan, not an authorization to operate high-voltage equipment.

7.2 Hypotheses and Testable Predictions

The preregistered hypotheses are:

1. Under a fixed geometry and a sufficiently slow, stable operating condition, the measured liquid interface will contain an approximately conical flank whose fitted half-angle can be compared with the simulated interface angle.
2. At matched geometry and fluid properties, the simulation will predict the direction and approximate magnitude of changes in interface shape when the electrode spacing or applied voltage is varied within the model's stated regime.
3. The prediction error will be distinguishable from image-processing and repeatability uncertainty. If it is not, the experiment will be reported as inconclusive rather than as validation.

The ideal 49.29° Taylor half-angle is a theoretical benchmark, not an acceptance value that every finite needle experiment must reproduce. Finite nozzle radius, finite electrode spacing, counter-electrode geometry, flow, conductivity, and the choice of where the flank is fitted can shift the measured angle. Published experimental and modeling studies show that electrode geometry, capillary diameter, voltage, flow rate, and conductivity influence electrospray behavior and onset conditions [7, 8, 9].

7.3 Variables and Observables

The primary response variables are the reconstructed interface profile $R_{\mathrm{exp}}(z)$, the flank half-angle α_{exp}, and the repeat-to-repeat variation of those quantities. The onset voltage V_{on} and qualitative operating state are secondary response variables. The operating-state labels will be defined before analysis as: no visible electrical deformation, deformed meniscus, unstable cone-like state, stable cone or cone–jet state, and post-onset spray or breakup. These labels are descriptive observations, not claims that the reduced model resolves all underlying dynamics.

Every trial will record the following variables:

Geometry. Record the nozzle material and dimensions, inner and outer diameters, exposed length, electrode type and dimensions, electrode spacing, axial alignment, radial offset, and the reference coordinate used for image–simulation alignment. The extractor geometry must be held fixed for the primary comparison because counter-electrode shape and spacing affect onset behavior [10].

Fluid. Record liquid identity, supplier and lot where available, composition, preparation date, temperature, surface tension γ, density ρ, dynamic viscosity μ, electrical conductivity σ, and relative permittivity ε_r. Values taken from literature must include the temperature and composition for which they apply; values that dominate the uncertainty should be measured or independently checked. Conductivity and flow rate are not incidental metadata: they alter the electrospray operating regime and can change the relation between cone shape and applied voltage [7, 6].

Operation. Record applied voltage and polarity, measured or independently verified voltage where permitted by the approved apparatus, flow rate, voltage step or ramp, dwell time, stabilization time after each change, and whether the run is an increasing or decreasing voltage sweep. The experiment should not use a nominal flow-rate range as a literature-derived guarantee of stability. Instead, the allowed flow range will be selected during supervised commissioning and reported with the resulting stability map. Cone–jet studies show that stable operation occupies a regime bounded by flow and electrical conditions rather than a universal flow interval [8, 9, 7].

Imaging. Record camera model, sensor resolution, frame rate, exposure, gain, lens, working distance, illumination, optical axis, field of view, calibration target, and any image-processing settings. The camera must view the cone approximately normal to its axis, and the optical arrangement must be checked for perspective and distortion. A planar calibration target and distortion model may be used following a calibrated-camera procedure such as Zhang's method [11]; a known needle diameter is a useful scale check but should not be the sole calibration if the needle is out of the focal plane.

Environmental and provenance metadata. Record date, time, ambient conditions when available, equipment identifiers, supervisor, operator, trial identifier, raw-video filename, analysis-code commit, and anomalies such as bubbles, wetting changes, contamination, vibration, discharge, or loss of alignment. Raw images must be retained and never overwritten by processed images.

7.4 Approved Apparatus Concept

The intended configuration is a liquid supplied through a nozzle facing a grounded counter-electrode, with optical imaging of the meniscus and cone. The final apparatus, voltage range, fluid, enclosure, interlocks, current limiting, discharge procedure, ventilation, and grounding scheme must be specified and approved by the supervisor before use. The paper will report the actual components and settings used, not the provisional examples in this planning document.

The primary geometry should use one nozzle and one counter-electrode configuration throughout the baseline trials. A second geometry may be added only after the baseline procedure is repeatable. Changing nozzle diameter, gap, extractor shape, fluid composition, or flow rate simultaneously would make the result non-identifiable because several mechanisms would change at once.

7.5 Commissioning and Calibration

Before collecting validation data, the team will perform a non-energized calibration and imaging check. The optical axis, field of view, focus, illumination uniformity, pixel scale, and distortion correction will be documented. Calibration should be performed in the same focal plane and with the same lens configuration used for the cone images. The scale will be checked against an independent dimension, such as the measured nozzle diameter, and the discrepancy will be included in the uncertainty budget.

The image-processing pipeline will first be tested on synthetic silhouettes with known angle, profile, blur, pixel scale, and noise. This test evaluates measurement bias only; it does not validate the physical solver. The acceptance limits for angle and profile recovery will be fixed before experimental images are processed, and the synthetic test set will not be tuned using the experimental frames.

7.6 Trial Procedure

For each approved trial, the operator will first confirm the configuration, fluid identification, calibration status, enclosure status, and data-recording path. The liquid supply and imaging system will be allowed to reach the documented initial condition. The system will then be observed at the approved baseline condition before any voltage sweep.

Voltage changes, dwell times, and any flow adjustments will follow the supervisor-approved operating procedure. Video will be recorded continuously or at a documented sampling schedule. After each change, the operator will record the applied condition and allow the prescribed stabilization interval before classifying the state. The classification will be based on predeclared image and observation criteria rather than on the desired outcome.

The voltage sweep will be performed in both increasing and, only if approved, decreasing directions so that possible hysteresis is visible. If the increasing and decreasing transitions differ, both values will be reported; they will not be silently averaged into a single onset value. At least three independent repeats should be attempted for the baseline condition when material and time allow, with the independence definition stated explicitly (for example, separate refill and reinitialization rather than many adjacent frames from one run).

The experiment will stop immediately under the supervisor-approved stop conditions, including any sign of electrical discharge, uncontrolled spray, leakage, overheating, loss of enclosure integrity, or other unsafe condition. These conditions are safety controls and are not scientific exclusion rules to be applied selectively after seeing the results.

7.7 Operational Definition of Experimental Onset

Because "onset" is used differently in the literature, this project will distinguish two observables. *Deformation onset* is the first voltage at which the meniscus departs from its baseline shape by a predeclared image-based criterion. *Cone–jet onset* is the first voltage at which a cone-like meniscus and sustained jet or spray satisfy a predeclared stability criterion. The paper will state which definition is used in each comparison. Studies of electrospray operation commonly identify multiple regimes and a finite stability region rather than a single universal threshold [8, 9, 7, 10].

A provisional stability rule may require a contiguous observation window in which the fitted flank angle and apex position remain within predeclared tolerances, with no visible breakup or discharge. The numerical tolerances and observation duration must be set during commissioning and frozen before the validation dataset is analyzed. They are operational measurement rules, not universal physical constants, and will be reported as such.

The experimental onset uncertainty will include voltage-step resolution, repeat-to-repeat variation, and any transition bandwidth. If the state changes within a voltage interval, the result will be reported as an interval rather than as an artificially precise single number.

7.8 Model–Experiment Matching

For each trial, the simulation input will be generated from the recorded metadata. The matched input must include the actual electrode geometry, gap, nozzle reference geometry, fluid surface tension, and any other parameters used by the model. The gas boundary conditions and immersed-interface conventions will be recorded with the solver commit. If a measured quantity is unavailable, the input will be reported as an assumption and varied in a sensitivity analysis rather than hidden.

The primary comparison will use the interface profile at a matched operating condition. If the solver predicts an onset amplitude rather than a dynamic cone–jet transition, that prediction will be compared with the measured electrostatic–capillary state, not labeled as a prediction of dynamic onset. The present model's grounded-box result and imposed-Taylor verification are numerical verification results; they do not, by themselves, establish agreement with a real finite-nozzle experiment.

The simulation prediction package will be frozen before image processing. It will contain the predicted profile, angle, model-defined voltage quantity, residual diagnostics, grid-resolution information, input metadata, solver commit, and a SHA-256 hash of the output file. Any later parameter change will be labeled post-validation analysis.

7.9 Image Analysis

The analysis will use raw frames, not screenshots or manually redrawn contours. The pipeline will: correct or document lens distortion; crop a fixed region of interest; apply a predeclared intensity normalization or denoising operation; detect the liquid–gas boundary; select the physically connected contour; convert pixels to (r, z) coordinates; and save both the extracted contour and processing parameters. A calibration target is preferred; the needle-diameter method is a secondary check.

The cone angle will be fitted only over a predeclared flank window that excludes the rounded apex, nozzle attachment, and any visible jet. The fitting convention must match the solver's convention. If $r = mz + b$ is fitted in coordinates where z is axial distance along the cone, then $\alpha_{\mathrm{exp}} = \arctan(m)$ after confirming that the selected coordinate orientation makes m positive. The angle uncertainty will combine regression uncertainty, frame-to-frame variation, calibration uncertainty, and sensitivity to the allowed flank window.

The two sides of a nominally axisymmetric silhouette will be analyzed separately before being averaged. Their difference is a useful diagnostic of optical misalignment, asymmetric wetting, vibration, or genuinely non-axisymmetric behavior. A trial that visibly violates axisymmetry will not be silently forced into an axisymmetric comparison; it will be flagged and analyzed separately.

7.10 Validation Metrics and Acceptance Logic

The primary profile metric is

$$\mathrm{RMSE}_R = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[R_{\mathrm{sim}}(z_i) - R_{\mathrm{exp}}(z_i)\right]^2}, \tag{66}$$

with a normalized value reported relative to a declared characteristic length L_c. The angle error and voltage error are

$$\Delta\alpha = |\alpha_{\text{sim}} - \alpha_{\text{exp}}|, \qquad \epsilon_V = \frac{|V_{\text{sim}} - V_{\text{exp}}|}{V_{\text{exp}}}. \tag{67}$$

No universal literature threshold will be assigned to these errors. Before data collection, the team will declare a measurement-based acceptance rule, for example: the model is supported for a metric if its prediction lies within the predeclared combined uncertainty interval and if the residual is smaller than a specified baseline model or calibration-error scale. The rule must be fixed before inspecting the experimental result. A prediction outside the interval is evidence of model discrepancy, not permission to retune the simulation.

Results will include repeat means, standard deviations or confidence intervals appropriate to the sample size, and an uncertainty budget separating calibration, contour extraction, frame selection, repeatability, and model-input uncertainty. Profile errors will also be reported separately for apex, flank, and base regions because the current model does not represent all three regions with equal fidelity.

7.11 What Would Count as Validation?

A successful experiment would provide evidence for the model only if: the apparatus and fluid properties are documented; the image pipeline passes the blinded synthetic test; repeated trials are stable enough to define uncertainty; the measured geometry lies within the frozen prediction interval at matched conditions; and the result is reproduced across at least one controlled change in geometry or voltage without refitting model parameters.

A result that reproduces an approximately conical silhouette but misses the profile, angle trend, or onset behavior would support only the qualitative Taylor-cone picture. A result that matches the half-angle at one condition but fails under a controlled geometry change would not validate the solver. Conversely, a mismatch in dynamic cone–jet onset would not necessarily falsify the static electrostatic–capillary model, because flow, conductivity, charge transport, space charge, and jet dynamics are outside its present scope. Those mechanisms would need to be measured or modeled before making a stronger claim.

The realistic outcome is therefore a tiered validation statement: (i) image-analysis validity, (ii) static interface-shape comparison, (iii) controlled geometry/voltage trend comparison, and only then (iv) limited onset-voltage comparison. The experiment has a credible possibility of validating the model's geometry predictions, but a substantially lower possibility of validating its full electrospray onset physics without extending the model to include the missing electrohydrodynamic mechanisms.

8 Image Processing and Comparison Method

8.1 Computer-Vision Pipeline

The image-processing pipeline will convert recorded Taylor-cone images into quantitative interface coordinates. The planned workflow is:

1. select stable frames from video sequences,
2. crop the region of interest around the meniscus,
3. calibrate pixels to physical length using a known needle/nozzle dimension or calibration target,
4. apply thresholding and/or edge detection,

5. extract the cone boundary contour,
6. fit local interface slopes near the cone region,
7. compute the experimental half-angle,
8. export $(r_{\text{exp}}, z_{\text{exp}})$ profile coordinates and uncertainty estimates.

Before physical data are collected, the pipeline will be tested on synthetic cone images with known ground-truth angles. The current end-to-end implementation has not passed this qualification: a diagnostic rerun on a nominal 49.29° cone produced large angle and profile errors even at zero added noise. Consequently, this section is a proposed qualification protocol only; it is not evidence that the pipeline, camera calibration, or physical solver is validated. The repair gate requires held-out rendered images, explicit failure counts, fixed coordinate conventions, and predeclared thresholds before any experimental accuracy claim is made.

8.1.1 Frame Selection

Video sequences recorded during voltage ramps will contain hundreds to thousands of frames. Frame selection identifies the subset of frames that satisfy the stability criteria defined in Section 7. The selection algorithm will:

1. Divide the video into segments corresponding to each voltage step (using timestamps or manual annotation).
2. For each segment, compute a time series of shape metrics (e.g., apex position, interface curvature, centroid location).
3. Identify the longest contiguous sub-interval within the segment where shape metrics remain within the stability tolerance bands.
4. Select representative frames from the stable interval. Extract each frame separately and report the within-interval variation; do not treat adjacent video frames as independent experimental trials.
5. Use a predeclared summary (for example, the median profile) only after checking that frame-to-frame variation is small relative to the measurement uncertainty.

If no stable interval exists within a voltage segment, that segment will be classified as "unstable" and excluded from quantitative comparison.

8.1.2 Calibration

Pixel-to-millimeter calibration will use one of two methods:

1. **Calibration target method:** A precision-machined scale or printed checkerboard pattern will be placed in the focal plane before the experiment. The pixel spacing corresponding to a known physical distance will be measured using edge detection or corner detection. This method is preferred when the camera-to-object distance is fixed and repeatable.
2. **Needle diameter method:** The outer diameter of the needle will be measured independently using digital calipers or microscopy (e.g., $d_{\text{out}} = 0.80 \pm 0.02$ for a standard 21-gauge needle). In the image, the needle diameter in pixels will be measured by fitting two parallel lines to the needle sidewalls far from the meniscus. The calibration scale is then $\text{mm/pixel} = d_{\text{out}}/d_{\text{pixels}}$.

Calibration uncertainty will be propagated through to all extracted dimensions. If lens distortion is significant, a distortion correction map will be applied using the calibrated-camera procedure

described by Zhang [11]. The distortion threshold, if one is used, will be declared from the calibration residuals rather than treated as a universal 1% rule.

8.1.3 Image Preprocessing and Edge Detection

Raw grayscale images will be preprocessed to enhance the liquid-gas interface contrast:

1. **Cropping:** Extract a region of interest (ROI) containing the needle tip and meniscus, discarding background pixels far from the interface.
2. **Histogram equalization or normalization:** Adjust intensity distribution to maximize dynamic range within the ROI.
3. **Gaussian blur (optional):** Apply a small Gaussian kernel (e.g., $\sigma = 1$–2 pixels) to reduce sensor noise, with kernel size chosen to preserve edge sharpness.

Edge detection will use one of the following methods, selected based on performance on synthetic validation images:

- **Canny edge detection:** A two-threshold gradient-based method that produces thin, connected edges. Hysteresis thresholds will be tuned on synthetic images and held fixed for all experimental trials.
- **Otsu thresholding + contour extraction:** Automatically select an intensity threshold that separates liquid (dark) from gas (bright) regions, then extract the boundary contour using a connected-component algorithm. This method is robust to lighting variation but may struggle with low-contrast or noisy images.
- **Sobel gradient magnitude + non-maximum suppression:** Compute the image gradient, threshold the magnitude, and suppress non-local-maximum pixels to produce thin edges. This is a simpler alternative to Canny.

The chosen method will be documented in the final validation report along with all parameter values (e.g., Canny thresholds, Gaussian σ, Otsu method).

8.1.4 Contour Extraction and Smoothing

After edge detection, the interface contour will be extracted as an ordered list of pixel coordinates $\{(x_i, y_i)\}_{i=1}^{N_{\text{contour}}}$. Post-processing steps include:

1. **Contour selection:** If multiple contours are detected (e.g., needle edge, interface, artifacts), select the contour that:
   - originates near the needle tip,
   - extends into the region between the needle and extractor,
   - has monotonically increasing axial coordinate (no backtracking),
   - has the longest arclength among candidates.
2. **Axisymmetry enforcement:** Since the experiment observes a 2D projection of an axisymmetric interface, the pipeline will extract one side of the contour (e.g., left or right boundary) and assume radial symmetry. If both sides are visible, the two half-profiles can be averaged to reduce asymmetry artifacts from optical misalignment.
3. **Smoothing (optional):** Apply a low-pass filter (e.g., Savitzky–Golay filter, cubic spline with smoothing parameter) to reduce pixel-level noise while preserving the overall cone shape. The smoothing strength will be validated on synthetic images to ensure it does not bias the extracted angle.

8.1.5 Coordinate Transformation to (r, z)

The pixel coordinates will be transformed into the solver's axisymmetric coordinate system:

1. Apply the calibration scale: $(x_{\text{mm}}, y_{\text{mm}}) = \text{scale} \times (x_{\text{px}}, y_{\text{px}})$.
2. Identify the needle tip or apex as the coordinate origin. For needle-tip referencing, locate the point where the interface detaches from the needle (sharp curvature change or minimum radius). For apex referencing, locate the point of maximum axial penetration into the gas.
3. Define the axial direction z (parallel to needle axis, positive toward extractor) and the radial direction r (perpendicular to axis, positive away from centerline).
4. Transform the contour into (r, z) coordinates: $r_i = |x_i|$ (distance from axis), $z_i = y_i$ (axial position).

The experimental profile is now represented as $\{(r_i^{\text{exp}}, z_i^{\text{exp}})\}_{i=1}^{N_{\text{exp}}}$.

8.1.6 Half-Angle Measurement

The cone half-angle will be extracted by fitting a straight line to the flank region of the experimental profile. The fitting procedure is:

1. **Define flank region:** Select the interval $z \in [z_{\min}, z_{\max}]$ where the interface is approximately conical. Exclude the apex region (high curvature, rounded cap) and the base region (near needle tip, attachment artifacts). A typical choice is the middle 50% of the visible cone length, e.g., $z_{\min} = z_{\text{apex}} - 0.25L_{\text{cone}}$, $z_{\max} = z_{\text{apex}} - 0.75L_{\text{cone}}$, where L_{cone} is the total cone length from apex to needle.
2. **Least-squares line fit:** Fit $r = mz + b$ to the flank points using weighted or unweighted least squares. The slope $m = \tan(\alpha)$ gives the half-angle:

$$\alpha_{\text{exp}} = \arctan(m). \tag{68}$$

3. **Uncertainty estimate:** The standard error of the slope σ_m from the regression can be propagated to the angle:

$$\sigma_\alpha = \frac{\sigma_m}{1 + m^2}. \tag{69}$$

   If multiple frames are analyzed, the angle standard deviation across frames provides an additional uncertainty estimate.

The flank-window selection will be validated on synthetic images to ensure it produces accurate angles for known cone geometries and is not biased by the window choice.

8.2 Profile Alignment

Simulation and experiment will be compared in the same coordinate system. Experimental pixel coordinates will be converted to millimeters and aligned so that the extracted apex or another reproducible reference point matches the simulation coordinate origin. The experimental radius $r_{\text{exp}}(z_i)$ will be interpolated onto the simulation z_i locations, or both profiles will be interpolated onto a common coordinate grid.

Alignment Strategy. Three alignment methods will be considered:

1. **Apex alignment:** Translate the experimental profile so that its apex (point of maximum z or minimum r) coincides with the simulated apex. This is geometrically natural but may be sensitive to apex-region noise or cap artifacts.

2. **Needle-tip alignment:** Translate both profiles so that the needle-tip detachment point (experimentally measured, simulated as the cone base) is at the origin. This is robust to apex uncertainty but requires clear identification of the detachment point in both experiment and simulation.
3. **Flank-region least-squares alignment:** Minimize the profile RMSE over translations and small rotations, using only the flank region (excluding apex and base). This is robust to localized artifacts but adds degrees of freedom that could artificially reduce error.

The primary method will be apex alignment unless preliminary validation on synthetic images shows it is unreliable. The alignment choice will be frozen before processing experimental data.

Interpolation. After alignment, one profile will be interpolated onto the other's coordinate grid. Linear or cubic spline interpolation will be used. If the experimental profile has higher spatial sampling than the simulation grid, the experimental points will be interpolated onto the simulation z_i locations. If the simulation is finer, the opposite will be done. Points where one profile extends beyond the other (e.g., experimental profile reaches closer to the base) will be excluded from RMSE calculation to avoid extrapolation artifacts.

8.3 Validation Metrics

The primary profile metric will be the root mean square error:

$$\mathrm{RMSE}_R = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_{\mathrm{sim}}(z_i) - R_{\mathrm{exp}}(z_i)\right)^2}. \tag{70}$$

A normalized version will divide by a characteristic length L_c, such as electrode spacing or observed cone length:

$$\mathrm{nRMSE}_R = \frac{\mathrm{RMSE}_R}{L_c}. \tag{71}$$

The cone-angle error will be

$$\Delta\alpha = |\alpha_{\mathrm{sim}} - \alpha_{\mathrm{exp}}|. \tag{72}$$

For onset voltage,

$$\epsilon_V = \frac{|V_{\mathrm{sim}} - V_{\mathrm{exp}}|}{V_{\mathrm{exp}}}. \tag{73}$$

Repeated trials will report mean, standard deviation, and uncertainty intervals where possible.

Profile RMSE Decomposition. To diagnose the source of profile mismatch, the radial error can be decomposed into components:

- **Angle-driven error:** If the simulated and experimental half-angles differ by $\Delta\alpha$, the radial profiles will diverge as $\Delta r \approx z\tan(\Delta\alpha)$ along the flank. This component can be estimated by comparing the RMSE from the actual profiles to the RMSE from two cones with half-angles α_{sim} and α_{exp} but otherwise identical geometry.
- **Apex-region error:** Near the apex, the experimental profile may have a finite-radius cap due to finite resolution or physical regularization (surface tension smoothing, finite needle radius), while the simulation may impose a sharper or differently sized cap. The apex-region contribution can be isolated by computing RMSE separately for $z < z_{\mathrm{apex}} - \delta$ (apex window) and $z > z_{\mathrm{apex}} - \delta$ (flank window).

- **Base-region error:** Near the needle attachment, contact-line dynamics, wetting behavior, and needle geometry artifacts may cause deviations not captured by the idealized cone model. Base-region RMSE can be computed separately.

Statistical Significance. Repeated trials at the same condition will produce a distribution of measured angles, onset voltages, and profile RMSEs. Agreement between simulation and experiment will be assessed by comparing the simulation prediction to the experimental mean ± standard deviation (or standard error of the mean if the trial count is small). If the prediction falls within the experimental uncertainty band, agreement is strong. If not, the discrepancy magnitude will be reported as a multiple of the experimental standard deviation (e.g., "simulation predicts $\alpha = 47.5°$, experiment measures $49.2 \pm 0.8°$, discrepancy is 2.1σ").

8.4 Synthetic Validation of the Image Processing Pipeline

Before analyzing experimental data, the entire image-processing pipeline will be validated on synthetic images generated from the solver output. The validation procedure is:

Step 1: Generate Synthetic Images.

1. Run the solver at a known geometry (e.g., electrode spacing $L = 10$, needle diameter $d = 0.8$, surface tension $\gamma = 0.022$, applied voltage $V = 3$).
2. Extract the predicted interface profile $R_{\text{sim}}(z)$ and half-angle α_{sim} (ground truth).
3. Render the profile as a 2D silhouette image using a graphics library (e.g., matplotlib filled contour, PIL polygon rendering, or OpenCV drawing functions).
4. Embed the silhouette in a realistic image background: add a needle representation, set pixel resolution (e.g., 1024×768), apply calibration scale (e.g., 50), render a backlight gradient.
5. Corrupt the synthetic image with realistic noise and blur: add Gaussian intensity noise (SNR 40–60 dB), apply Gaussian blur ($\sigma = 1$–2 pixels), simulate lens distortion if applicable, add pixel quantization (8-bit or 12-bit depth).

Step 2: Process Synthetic Images Through Pipeline.

1. Apply the full image-processing pipeline (edge detection, contour extraction, calibration, angle measurement) to the synthetic image.
2. Extract $\alpha_{\text{extracted}}$ and $R_{\text{extracted}}(z)$ from the pipeline output.
3. Compute the round-trip errors:

$$\Delta\alpha_{\text{synthetic}} = |\alpha_{\text{extracted}} - \alpha_{\text{sim}}|, \tag{74}$$

$$\text{RMSE}_{\text{synthetic}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_{\text{extracted}}(z_i) - R_{\text{sim}}(z_i)\right)^2}. \tag{75}$$

Step 3: Acceptance Criteria. The image-processing pipeline will be accepted as valid if the synthetic validation shows:

- $\Delta\alpha_{\text{synthetic}} < 1°$ for cone angles in the range $40°$ to $55°$ under moderate noise (SNR ≥ 40 dB),
- $\text{nRMSE}_{\text{synthetic}} < 2\%$ of the cone length,
- Round-trip error does not increase systematically with angle, cone length, or noise level (indicating method robustness).

If this future synthetic qualification fails its predeclared criteria, the edge-detection method, smoothing parameters, or flank-window selection will be revised before experimental images are analyzed. Until that gate passes, no synthetic image result is presented as evidence that the image-analysis method is unbiased, experimentally calibrated, or physically validated.

8.5 Planned Output Figures

The final paper will include: raw image frames, detected-edge overlays, extracted cone profiles, simulated-versus-experimental profile overlays, half-angle comparison plots, onset-voltage comparison plots, and runtime scaling figures.

Representative Image Sequence. A figure will show 3–5 frames from a single voltage ramp: (a) no cone (hemispherical meniscus at low voltage), (b) deformed meniscus (subcritical), (c) stable Taylor cone at onset, (d) cone with incipient jet or spray (post-onset). Each frame will include the detected edge overlay and annotated voltage.

Profile Comparison. For each validated trial, a plot will overlay the experimental profile $R_{\text{exp}}(z)$ (solid line with error band) and the frozen simulation prediction $R_{\text{sim}}(z)$ (dashed line) in the same coordinate system. The half-angles will be annotated with fitted flank lines. Residual error $R_{\text{sim}}(z) - R_{\text{exp}}(z)$ will be plotted as a function of z in a separate panel.

Angle and Voltage Comparison. Scatter plots or bar charts will compare simulated versus experimental half-angles and onset voltages across all trials. Error bars will represent experimental uncertainty (standard deviation across repeated trials). A diagonal line $y = x$ will indicate perfect agreement. The plots will quantify systematic bias (mean offset) and random scatter (RMSE about the diagonal).

Voltage-Sweep Landscape. For at least one trial, a figure will show the experimental regime map (voltage vs qualitative state: no cone, unstable, stable, spray) overlaid with the simulated residual landscape (residual RMS vs voltage). This will illustrate how the numerical onset criterion aligns with the experimental onset observation.

Synthetic Validation Results. A supplementary figure will document the synthetic validation: (a) example synthetic image with ground-truth profile overlaid, (b) extracted profile from pipeline, (c) angle error and profile RMSE as functions of noise level and cone angle, demonstrating pipeline accuracy and robustness.

9 Expected Results and Experimental Placeholders

The current submission draft is prepared before physical experiments have been completed. Therefore, experimental result locations are explicitly marked as placeholders rather than fabricated or assumed data.

9.1 Computational Results Already Available

Current completed computational results include:

- classical Taylor half-angle benchmark: 49.2900892921°,

- manufactured Poisson convergence under grid refinement,
- voltage scaling check showing $E \propto V$ and $p_E \propto V^2$,
- residual null sanity check with near-zero RMS residual,
- Gaussian and threshold space-charge closure demonstrations,
- automated test suite passing all current tests.

9.2 Experimental Tables to Complete

Table 2: Placeholder for experimental half-angle comparison.

Trial	V_{exp}	α_{exp}	α_{sim}	$\Delta\alpha$
1	TBD	TBD	TBD	TBD
2	TBD	TBD	TBD	TBD
3	TBD	TBD	TBD	TBD

Table 3: Placeholder for interface-profile and onset-voltage validation.

Condition	Profile RMSE	nRMSE	V_{exp}	ϵ_V
Baseline	TBD	TBD	TBD	TBD
Spacing sweep 1	TBD	TBD	TBD	TBD
Spacing sweep 2	TBD	TBD	TBD	TBD

9.3 Criteria for Interpreting Agreement

Agreement will not be treated as binary proof. Instead, the project will report quantitative error and compare it with experimental uncertainty and model assumptions. Strong agreement would mean small angle error, low normalized profile RMSE, and correct onset-voltage trends across controlled geometry changes. Disagreement will be analyzed to identify whether the dominant cause is experimental uncertainty, finite-grid geometry, simplified space-charge closure, or missing physics.

10 Safety, Ethics, and Compliance

The project contains a computational component and a proposed experimental validation component. The computational component can proceed independently. The physical experiment will not begin until appropriate school approval, local scientific review, risk assessment, and qualified adult supervision are secured.

The proposed experimental validation may involve high voltage and flammable solvent, so it must be treated as a hazardous-device and hazardous-chemical activity. Before purchasing, assembling, or operating any apparatus, the project team will prepare required approval forms, a full risk assessment, safety data sheet documentation, supervision plan, schematic review, operating boundaries, and emergency shutoff procedures. Operation will occur only in a supervised school or laboratory environment, with appropriate enclosure, interlock, current limiting, grounding, ventilation, PPE, and waste-disposal procedures as required by school and competition rules.

No live high-voltage demonstration is planned for a public fair setting. Experimental data, images, videos, and analysis results will be presented instead. The apparatus is intended for microfluidic Taylor-cone observation and image-based validation, not as a projectile device or propulsion demonstration. The project will comply with all applicable Hong Kong selection-fair and Regeneron ISEF-affiliated rules.

11 Limitations

The present solver is intentionally reduced-order. It does not solve full transient Navier–Stokes equations, full Taylor–Melcher leaky-dielectric surface charge conservation, droplet breakup, plume dynamics, or complete thruster performance. The effective space-charge closures are phenomenological models for studying Poisson shielding trends, not predictive emitted-current laws.

The axisymmetric assumption is appropriate for onset-level symmetric geometries but may fail after non-axisymmetric instabilities or downstream jet breakup appear. The finite-difference grid representation introduces geometric error, especially near sharp conical interfaces. The current optimizer-angle deviation from the ideal Taylor half-angle highlights this limitation and motivates improved boundary treatment or explicit grid-convergence quantification.

Experimental validation will also have limitations. Cone-angle extraction depends on image resolution, optical alignment, lighting, threshold selection, calibration accuracy, and frame selection. Onset voltage may depend on fluid impurities, humidity, flow stability, electrode condition, and subjective stability criteria unless carefully standardized. Therefore, final conclusions will be stated in terms of quantified agreement and uncertainty rather than exact prediction.

12 Conclusion and Future Work

This project develops a lightweight sparse Python solver for onset-level Taylor-cone modeling. The current codebase already includes an axisymmetric electrostatic solver, field reconstruction, residual diagnostics, effective space-charge closures, a shape optimizer, a Streamlit app, automated tests, and numerical validation artifacts. It reproduces the classical Taylor half-angle benchmark and passes current numerical sanity checks.

The next stage is experimental validation. After required approval and supervision, Taylor-cone images will be collected, processed using computer vision, and compared against solver predictions using profile RMSE, angle error, onset-voltage error, and runtime metrics. This will test whether the reduced-order model is accurate enough for accessible early-stage parameter studies on consumer hardware.

Future work includes improving sharp-interface boundary representation, coupling threshold space charge inside the shape-optimization loop, adding more physically constrained charge/current closures, extending toward leaky-dielectric effects where justified, and comparing against higher-fidelity simulations or published experimental datasets. The intended contribution is not to replace full multiphysics simulation, but to provide a transparent, reproducible, and computationally accessible model for Taylor-cone onset studies.

References

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A Submission Checklist

Before final submission, the following items should be completed:

- Confirm official Hong Kong selection-fair formatting requirements.
- Replace placeholder author/school information.
- Confirm word/page limits and adjust length if needed.
- Add final experimental figures and tables after approved data collection.
- Add uncertainty estimates and repeated-trial statistics.
- Confirm that all safety approvals are documented before experimental work.
- Clean repository paths and ensure reproducible commands are correct.
- Verify all citations and bibliography formatting.
- Compile final PDF and check figures, tables, hyperlinks, and equation numbering.

B Reproducibility Notes

The solver repository is organized so that numerical results can be reproduced from the project root. The current local path is:

```
/Users/a1/Desktop/isef
```

The Python environment used during this draft is Miniforge Python:

```
/Users/a1/miniforge3/bin/python
```

Important commands are:

```
/Users/a1/miniforge3/bin/python -m pytest -q
/Users/a1/miniforge3/bin/python examples/04_taylor_angle_benchmark.py
/Users/a1/miniforge3/bin/python examples/05_shape_optimization_demo.py
/Users/a1/miniforge3/bin/python -m streamlit run app/streamlit_app.py
```

The final repository should include scripts for exporting simulation profiles, processing experimental images, and generating comparison figures.